

### 3. Run Apriori algorithm to find frequent item sets and association rules on 2 real datasets and use appropriate evaluation measures to compute correctness of obtained patterns

- a) Use minimum support as 50% and minimum confidence as 75%
- b) Use minimum support as 60% and minimum confidence as 60 %

```
In [1]: from efficient_apriori import apriori
import pandas as pd
import numpy as np

groceries = pd.read_csv("groceries.csv", header=0)
print(groceries.head())
print(groceries.shape)

bakery = pd.read_csv("bakery.csv")
bakery.columns = bakery.columns.str.strip()
print(bakery.head())
print(bakery.shape)
```

|   | Item(s) | Item 1           | Item 2              | Item 3 \       |
|---|---------|------------------|---------------------|----------------|
| 0 | 4       | citrus fruit     | semi-finished bread | margarine      |
| 1 | 3       | tropical fruit   | yogurt              | coffee         |
| 2 | 1       | whole milk       | NaN                 | NaN            |
| 3 | 4       | pip fruit        | yogurt              | cream cheese   |
| 4 | 4       | other vegetables | whole milk          | condensed milk |

|   |                          | Item 4       | Item 5 | Item 6 | Item 7 | Item 8 | Item 9 | ... Item 23 \ |
|---|--------------------------|--------------|--------|--------|--------|--------|--------|---------------|
| 0 |                          | ready soups  | NaN    | NaN    | NaN    | NaN    | NaN    | NaN           |
| 1 |                          | NaN          | NaN    | NaN    | NaN    | NaN    | NaN    | NaN           |
| 2 |                          | NaN          | NaN    | NaN    | NaN    | NaN    | NaN    | NaN           |
| 3 |                          | meat spreads | NaN    | NaN    | NaN    | NaN    | NaN    | NaN           |
| 4 | long life bakery product | NaN          | NaN    | NaN    | NaN    | NaN    | NaN    | NaN           |

|   | Item 24 | Item 25 | Item 26 | Item 27 | Item 28 | Item 29 | Item 30 | Item 31 | Item 32 |
|---|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 0 | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     |
| 1 | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     |
| 2 | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     |
| 3 | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     |
| 4 | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     | NaN     |

[5 rows x 33 columns]  
(9835, 33)

|   | ID | Chocolate Cake | Lemon Cake | Casino Cake | Opera Cake | Strawberry Cake \ |
|---|----|----------------|------------|-------------|------------|-------------------|
| 0 | 1  | 0              | 0          | 0           | 0          | 1                 |
| 1 | 2  | 0              | 1          | 0           | 0          | 0                 |
| 2 | 3  | 0              | 0          | 0           | 0          | 0                 |
| 3 | 4  | 0              | 0          | 0           | 0          | 0                 |
| 4 | 5  | 0              | 0          | 0           | 0          | 0                 |

|   | Truffle Cake | Chocolate Eclair | Coffee Eclair | Vanilla Eclair | ... \ |
|---|--------------|------------------|---------------|----------------|-------|
| 0 | 1            | 1                | 0             | 0              | ...   |
| 1 | 0            | 0                | 0             | 0              | ...   |
| 2 | 0            | 0                | 0             | 0              | ...   |
| 3 | 0            | 0                | 0             | 0              | ...   |
| 4 | 0            | 0                | 0             | 0              | ...   |

|   | Lemon Lemonade | Raspberry Lemonade | Orange Juice | Green Tea | Bottled Water \ |
|---|----------------|--------------------|--------------|-----------|-----------------|
| 0 | 0              | 0                  | 0            | 0         | 0               |
| 1 | 0              | 0                  | 0            | 0         | 0               |
| 2 | 0              | 0                  | 0            | 0         | 0               |
| 3 | 0              | 0                  | 0            | 0         | 0               |
| 4 | 0              | 0                  | 0            | 0         | 0               |

|   | Hot Coffee | Chocolate Coffee | Vanilla Frappuccino | Cherry Soda \ |
|---|------------|------------------|---------------------|---------------|
| 0 | 0          | 0                | 0                   | 0             |
| 1 | 0          | 1                | 1                   | 0             |
| 2 | 0          | 0                | 0                   | 0             |
| 3 | 0          | 0                | 0                   | 0             |
| 4 | 0          | 0                | 0                   | 0             |

|   | Single Espresso |
|---|-----------------|
| 0 | 0               |
| 1 | 0               |
| 2 | 0               |
| 3 | 0               |

[5 rows x 51 columns]  
(5000, 51)

```
In [2]: transactions = []
        for i in range(len(groceries)):
            row = groceries.iloc[i, 1:].dropna().astype(str).tolist()
            transactions.append(row)

        print(len(transactions), transactions[:5])

        transactions_b = []
        item_columns = bakery.columns[1:]
        for i in range(len(bakery)):
            items = item_columns[bakery.iloc[i, 1:] == 1].tolist()
            if items:
                transactions_b.append(items)

        print(len(transactions_b), transactions_b[:5])
```

9835 [['citrus fruit', 'semi-finished bread', 'margarine', 'ready soups'], ['tropical fruit', 'yogurt', 'coffee'], ['whole milk'], ['pip fruit', 'yogurt', 'cream cheese', 'meat spreads'], ['other vegetables', 'whole milk', 'condensed milk', 'long life bakery product']]

5000 [['Strawberry Cake', 'Truffle Cake', 'Chocolate Eclair', 'Almond Tart'], ['Lemon Cake', 'Apple Tart', 'Lemon Tart', 'Chocolate Coffee', 'Vanilla Frappuccino'], ['Blueberry Tart', 'Apricot Croissant'], ['Cherry Tart', 'Lemon Cookie', 'Apricot Danish'], ['Apple Tart', 'Gongolais Cookie', 'Marzipan Cookie', 'Almond Croissant']]

```
In [3]: itemsets, rules = apriori(
        transactions,
        min_support=0.05,
        min_confidence=0.25
    )

    print("Frequent Itemsets (Groceries):")
    print(itemsets)
    print("\nRules (Groceries):")
    print(rules)

    b_item, b_rules = apriori(
        transactions_b,
        min_support=0.05,
        min_confidence=0.25
    )

    print("\nFrequent Itemsets (Bakery):")
    print(b_item)
    print("\nRules (Bakery):")
    print(b_rules)
```

#### Frequent Itemsets (Groceries):

{1: {'citrus fruit',): 814, ('margarine',): 576, ('tropical fruit',): 1032, ('yogurt',): 1372, ('coffee',): 571, ('whole milk',): 2513, ('pip fruit',): 744, ('other vegetables',): 1903, ('butter',): 545, ('rolls/buns',): 1809, ('bottled beer',): 792, ('bottled water',): 1087, ('curd',): 524, ('beef',): 516, ('frankfurter',): 580, ('soda',): 1715, ('fruit/vegetable juice',): 711, ('newspapers',): 785, ('pastry',): 875, ('root vegetables',): 1072, ('canned beer',): 764, ('sausage',): 924, ('brown bread',): 638, ('shopping bags',): 969, ('napkins',): 515, ('pork',): 567, ('whipped/sour cream',): 705, ('domestic eggs',): 624}, 2: {'other vegetables', 'whole milk': 736, ('rolls/buns', 'whole milk': 557, ('whole milk', 'yogurt'): 551}}

#### Rules (Groceries):

[{whole milk} -> {other vegetables}, {other vegetables} -> {whole milk}, {rolls/buns} -> {whole milk}, {yogurt} -> {whole milk}]

#### Frequent Itemsets (Bakery):

{1: {'Strawberry Cake',): 480, ('Truffle Cake',): 438, ('Lemon Cake',): 425, ('Apple Tart',): 367, ('Lemon Tart',): 354, ('Chocolate Coffee',): 404, ('Vanilla Frappuccino',): 367, ('Blueberry Tart',): 426, ('Apricot Croissant',): 411, ('Cherry Tart',): 460, ('Lemon Cookie',): 321, ('Apricot Danish',): 446, ('Gongolais Cookie',): 477, ('Marzipan Cookie',): 418, ('Walnut Cookie',): 353, ('Chocolate Cake',): 399, ('Casino Cake',): 373, ('Berry Tart',): 391, ('Lemon Lemonade',): 324, ('Bottled Water',): 386, ('Opera Cake',): 418, ('Tuile Cookie',): 499, ('Apple Croissant',): 371, ('Apple Danish',): 391, ('Cherry Soda',): 335, ('Chocolate Tart',): 381, ('Coffee Eclair',): 554, ('Apple Pie',): 391, ('Almond Twist',): 407, ('Orange Juice',): 461, ('Raspberry Lemonade',): 339, ('Napolean Cake',): 409, ('Hot Coffee',): 513, ('Raspberry Cookie',): 320, ('Green Tea',): 310, ('Cheese Croissant',): 385, ('Blackberry Tart',): 380, ('Single Espresso',): 327}, 2: {'Apricot Danish', 'Cherry Tart': 256}}

#### Rules (Bakery):

[{Cherry Tart} -> {Apricot Danish}, {Apricot Danish} -> {Cherry Tart}]